

# **PRADED, Czech Republic**

WMO: 117350

Lat: 50.0829N

Lon: 17.2299E

Elev: 4895

StdP: 12.28

Time Zone: 1.00 (EUC)

Period: 86-97

WBAN: 99999

## **Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -0.5       | 3.7 | -8.8                          | 4.1 | 8.3  | -4.2 | 5.3 | 9.5  | 55.9                  | 21.9 | 49.4 | 24.6 | 20.6                  | 100  | 0.988 |

## **Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 9.8                    | 66.5            | 55.9 | 63.6 | 54.7 | 60.9 | 53.6 | 57.9                | 63.4 | 56.4 | 61.3 | 54.8 | 59.2 | 11.0                 | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 55.9                            | 79.9 | 59.5 | 54.3 | 75.4 | 58.1 | 52.8 | 71.3 | 56.6 | 27.2          | 64.2 | 26.1 | 61.4 | 25.0 | 59.3 | 64.8              |

## **Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 45.1          | 40.1 | 35.6 | DB                         | -4.3 | 70.9               | 8.3 | 2.1                                                | -10.3 | 72.5       | -15.1 | 73.7       | -19.8 | 74.9       | -25.8 | 76.5 | (4) |
| (5)               |      |      | WB                         | -4.8 | 61.4               | 8.1 | 1.8                                                | -10.6 | 62.7       | -15.3 | 63.8       | -19.8 | 64.9       | -25.7 | 66.2 | (5) |

## **Monthly Climatic Design Conditions**

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 34.8   | 21.6  | 21.0  | 24.2  | 31.7  | 41.8  | 46.5  | 50.7  | 50.9  | 42.6  | 36.7  | 27.1  | 22.1 |
|                                                                               | DBStd       | 13.44  | 8.22  | 9.14  | 8.31  | 8.48  | 7.30  | 6.69  | 6.54  | 7.07  | 6.37  | 8.09  | 7.10  | 7.76 |
|                                                                               | HDD50       | 5815   | 881   | 812   | 801   | 549   | 268   | 148   | 73    | 79    | 235   | 416   | 687   | 866  |
|                                                                               | HDD65       | 11020  | 1346  | 1232  | 1266  | 999   | 720   | 556   | 445   | 439   | 671   | 878   | 1137  | 1331 |
|                                                                               | CDD50       | 270    | 0     | 0     | 0     | 0     | 13    | 42    | 94    | 105   | 13    | 3     | 0     | 0    |
|                                                                               | CDD65       | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 0     | 0     | 0     | 0    |
|                                                                               | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 17.2   | 20.4  | 19.4  | 19.2  | 16.3  | 14.9  | 14.3  | 13.5  | 13.8  | 17.0  | 18.8  | 18.7  | 20.1 |
| Precipitation                                                                 | PrecAvg     | 34.6   | 2.4   | 2.1   | 2.5   | 2.2   | 3.3   | 4.0   | 4.7   | 3.4   | 3.1   | 2.3   | 2.4   | 2.4  |
|                                                                               | PrecMax     | 47.8   | 5.0   | 4.4   | 5.9   | 3.5   | 8.5   | 7.2   | 19.9  | 7.8   | 7.1   | 6.5   | 3.9   | 4.5  |
|                                                                               | PrecMin     | 23.4   | 1.0   | 0.5   | 0.9   | 0.2   | 1.3   | 1.5   | 1.3   | 0.9   | 0.7   | 0.3   | 0.0   | 0.7  |
|                                                                               | PrecStd     | 4.9    | 1.1   | 1.0   | 1.1   | 0.9   | 1.5   | 1.4   | 3.5   | 1.6   | 1.8   | 1.4   | 0.9   | 1.0  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 41.3  | 43.3  | 44.7  | 54.6  | 61.8  | 68.0  | 71.2  | 71.5  | 60.6  | 58.4  | 44.2  | 41.9 |
|                                                                               |             | MCWB   | 31.7  | 34.5  | 38.5  | 44.4  | 51.9  | 58.6  | 56.9  | 56.8  | 53.5  | 45.6  | 37.4  | 31.0 |
|                                                                               | 2%          | DB     | 36.0  | 39.5  | 40.7  | 50.7  | 57.4  | 62.9  | 66.4  | 66.7  | 56.3  | 53.0  | 41.1  | 37.7 |
|                                                                               |             | MCWB   | 31.3  | 32.2  | 35.5  | 42.0  | 49.0  | 55.6  | 55.3  | 56.2  | 52.0  | 44.7  | 38.3  | 33.0 |
|                                                                               | 5%          | DB     | 33.6  | 35.8  | 37.8  | 47.4  | 54.6  | 59.5  | 63.3  | 63.9  | 53.8  | 49.5  | 39.0  | 35.0 |
|                                                                               |             | MCWB   | 31.3  | 31.0  | 35.1  | 41.0  | 46.8  | 53.3  | 54.2  | 55.1  | 50.9  | 44.9  | 36.8  | 31.6 |
|                                                                               | 10%         | DB     | 31.7  | 32.6  | 35.0  | 43.7  | 51.9  | 56.6  | 60.3  | 61.0  | 51.9  | 47.0  | 36.7  | 32.5 |
|                                                                               |             | MCWB   | 29.9  | 30.6  | 33.2  | 38.9  | 46.2  | 51.6  | 53.4  | 53.6  | 49.9  | 44.1  | 34.8  | 30.6 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 34.9  | 36.8  | 40.7  | 46.5  | 53.6  | 59.4  | 60.2  | 59.5  | 55.8  | 50.2  | 41.1  | 37.5 |
|                                                                               |             | MCDB   | 35.8  | 40.7  | 42.2  | 50.3  | 58.7  | 64.2  | 65.3  | 66.2  | 57.8  | 52.8  | 41.9  | 38.2 |
|                                                                               | 2%          | WB     | 33.3  | 34.1  | 37.8  | 43.5  | 51.4  | 56.9  | 57.6  | 57.8  | 53.5  | 48.0  | 39.6  | 34.8 |
|                                                                               |             | MCDB   | 34.3  | 36.2  | 39.0  | 49.0  | 55.6  | 61.5  | 62.6  | 64.1  | 54.9  | 49.8  | 40.3  | 35.5 |
|                                                                               | 5%          | WB     | 31.9  | 32.3  | 35.4  | 41.3  | 49.3  | 54.7  | 55.9  | 56.4  | 51.9  | 46.4  | 37.8  | 32.6 |
|                                                                               |             | MCDB   | 33.3  | 33.8  | 36.7  | 46.4  | 52.5  | 58.5  | 60.8  | 61.8  | 53.3  | 48.4  | 38.4  | 33.4 |
|                                                                               | 10%         | WB     | 30.2  | 30.6  | 33.1  | 39.4  | 46.9  | 52.4  | 54.2  | 54.8  | 49.9  | 44.7  | 35.1  | 30.4 |
|                                                                               |             | MCDB   | 31.6  | 32.7  | 34.6  | 43.3  | 50.2  | 55.5  | 59.1  | 59.6  | 51.8  | 46.8  | 35.9  | 32.0 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 7.6   | 7.8   | 7.6   | 8.4   | 9.6   | 9.2   | 10.0  | 9.8   | 7.8   | 7.6   | 7.0   | 7.2  |
|                                                                               |             | MCDBR  | 8.3   | 10.9  | 10.9  | 11.5  | 12.5  | 11.9  | 12.9  | 12.6  | 10.3  | 10.7  | 9.6   | 8.5  |
|                                                                               |             | MCWBR  | 6.9   | 7.9   | 8.9   | 6.9   | 7.5   | 7.7   | 8.2   | 7.4   | 7.6   | 7.2   | 7.9   | 6.8  |
|                                                                               | 5% WB       | MCDBR  | 7.8   | 8.5   | 9.8   | 10.8  | 10.9  | 11.4  | 11.2  | 11.6  | 9.5   | 8.5   | 8.4   | 7.8  |
|                                                                               |             | MCWBR  | 7.3   | 7.4   | 9.2   | 7.3   | 7.8   | 8.2   | 8.2   | 7.9   | 8.2   | 7.1   | 7.7   | 7.1  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.253  | 0.273 | 0.303 | 0.339 | 0.339 | 0.339 | 0.350 | 0.345 | 0.321 | 0.301 | 0.282 | 0.253 |      |
|                                                                               | taud        | 2.409  | 2.414 | 2.410 | 2.366 | 2.386 | 2.400 | 2.364 | 2.392 | 2.453 | 2.496 | 2.501 | 2.458 |      |
|                                                                               | Ebn at Noon | 258    | 280   | 289   | 287   | 291   | 291   | 285   | 282   | 279   | 266   | 245   | 243   |      |
|                                                                               | Edh at Noon | 19     | 25    | 30    | 35    | 36    | 36    | 36    | 34    | 29    | 23    | 18    | 16    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 266    | 504   | 844   | 1317  | 1590  | 1689  | 1609  | 1445  | 1021  | 594   | 291   | 201   |      |
|                                                                               | RadStd      | 47     | 74    | 139   | 165   | 191   | 157   | 180   | 133   | 149   | 89    | 37    | 22    |      |

## **Historical Trends**

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (44 neighbors) | +0.90 | N/A     | N/A    | +1.37   | +0.65 | +0.56 | N/A         | -274  | +117  | +49   |

Nomenclature: See separate page